AWS EC2 Upgrade & Deployment Guide — Global ERP

Date: 2026-03-31 **Scope:** Upgrade existing EC2 instance + deploy last 3 days of changes + run 14 DB migrations **Assumption:** EC2 already running with Docker, Nginx, SSL, repo cloned, env files configured

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1. Changes Summary (Last 3 Days)

Commits (Mar 28 → Mar 31)

Commit	Description
`93ea99a`	Revenue Command Center (RCC) — AI engine, lead sources, pipeline, leakage, marketing spend
`8a93ff2`	Add `tmp/*.log` to `.gitignore`
`7c320b8`	Test flows — PIS advisor portal, test panel, vendor bids, inspection print, invoice pay, GRN PDF
`67719ba`	Merge remote main
`09e6062`	Reflect real data — PIS config, advisor scores, lead distribution, commission, SLA
`457d208`	Rename call center → sales center
`961f747`	Fix call center items
`f515564`	Fix recording issue
`c59be47`	Fix AIPanel auth + multi-provider AI config + e8 engine
`265d23b`	Call Center Performance Dashboard + e8 AI engine
`e62a79c`	Push
`42f3732`	Minor changes

Key Features

- **Revenue Command Center (RCC)** — 6 new pages + API routes + AI signal generator

- **PIS (Performance Incentive System)** — Advisor portal, lead distribution, commission, SLA
- **Multi-provider AI** — OpenAI + Anthropic dual provider
- **Call Center → Sales Center** rename
- **Test Panel** for flow testing
- **Vendor part details & bids** improvements
- **Service charges + main warehouse config**

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2. New Database Migrations

14 new migrations (173 → 186):

#	File	What it does
173	`173_multi_provider_ai_config.sql`	Multi-provider AI config table
174	`174_pis_config.sql`	PIS system config
175	`175_pis_advisor_scores.sql`	Advisor scoring tables
176	`176_pis_lead_distribution.sql`	Lead distribution rules
177	`177_pis_commission.sql`	Commission structure
178	`178_pis_sla_snapshots.sql`	SLA snapshot tracking
179	`179_pis_permissions.sql`	PIS RBAC permissions
180	`180_customers_insurance_name.sql`	Customer insurance name field
181	`181_advisor_portal_permission.sql`	Advisor portal RBAC
182	`182_vendor_part_details.sql`	Vendor part details schema
183	`183_main_warehouse_config.sql`	Main warehouse config
184	`184_service_charges_config.sql`	Service charges config
185	`185_rcc_marketing_spend.sql`	RCC marketing spend table
186	`186_rcc_permissions.sql`	RCC permissions

> Migration runner auto-skips already-applied migrations via `schema_migrations` table.

Part A: Upgrade EC2 Instance

A1. Check Current Instance & Decide Target Size

SSH into EC2:

```
ssh -i your-key.pem ubuntu@YOUR_EC2_IP
```

Check current resources:

```
# CPU & RAM
nproc
free -h

# Disk
df -h /

# OS version
lsb_release -a

# Docker version
docker --version
docker compose version

# Running containers
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
docker stats --no-stream
```

Recommended EC2 instance types:

Instance	vCPU	RAM	Use Case
`t3.medium`	2	4 GB	**Minimum** — tight for running dev+prod simultaneously
`t3.large`	2	8 GB	**Recommended** — comfortable for dev+prod+builds
`t3.xlarge`	4	16 GB	**Ideal** — fast builds, headroom for AI features

> With 14 new migrations, RCC module, PIS system, and AI multi-provider — **t3.large (8GB)** is the sweet spot.

A2. Resize EC2 Instance (if needed)

> Only needed if current instance is too small (e.g., t3.micro/small).

Via AWS Console:

1. Go to **EC2** → **Instances** → **Select your instance**
2. Click **Instance state** → **Stop instance** (wait for "Stopped")
3. Click **Actions** → **Instance settings** → **Change instance type**
4. Select `t3.large` (or your target)
5. Click **Apply**
6. Click **Instance state** → **Start instance**
7. Wait for status checks to pass

Via AWS CLI:

```
# From your local machine (not the EC2)
INSTANCE_ID="i-0xxxxxxxxxxx"
```

```
# Stop
aws ec2 stop-instances --instance-ids $INSTANCE_ID
aws ec2 wait instance-stopped --instance-ids $INSTANCE_ID

# Resize
aws ec2 modify-instance-attribute --instance-id $INSTANCE_ID --instance-type
'{"Value":"t3.large"}'

# Start
aws ec2 start-instances --instance-ids $INSTANCE_ID
aws ec2 wait instance-running --instance-ids $INSTANCE_ID

# Get new public IP (if using non-Elastic IP)
aws ec2 describe-instances --instance-ids $INSTANCE_ID \
  --query 'Reservations[0].Instances[0].PublicIpAddress' --output text
```

> **If you use an Elastic IP**, the IP stays the same after restart. If not, the public IP will change — update your DNS records.

A3. Expand EBS Volume (if disk is tight)

Check current disk:

```
df -h /
# If less than 20GB free, expand
```

Expand via AWS Console:

1. Go to **EC2** → **Volumes** (or find volume from instance details)
2. Select the root volume → **Actions** → **Modify volume**
3. Increase size (e.g., 40GB → 80GB)
4. Click **Modify**

Then SSH in and grow the filesystem:

```
# Check partition name
lsblk

# Grow partition (usually /dev/xvda1 or /dev/nvme0n1p1)
sudo growpart /dev/xvda 1
# OR for nitro instances:
sudo growpart /dev/nvme0n1 1

# Resize filesystem
sudo resize2fs /dev/xvda1
# OR for XFS:
sudo xfs_growfs /
```

```
# Verify
df -h /
```

A4. Update System & Docker on EC2

```
# SSH into EC2
ssh -i your-key.pem ubuntu@YOUR_EC2_IP

# Update OS packages
sudo apt update && sudo apt upgrade -y
sudo apt autoremove -y

# Upgrade Docker if needed (should be 24+)
docker --version
# If outdated:
curl -fsSL https://get.docker.com | sh

# Verify Docker Compose V2
docker compose version
```

A5. Update Security Group (if needed)

Ensure these inbound rules exist in your EC2 Security Group:

Port	Protocol	Source	Purpose
22	TCP	Your IP	SSH
80	TCP	0.0.0.0/0	HTTP (Nginx redirect)
443	TCP	0.0.0.0/0	HTTPS (Nginx + SSL)

> Ports 3000, 3001, 5432 should **NOT** be open to the internet — they're internal only (Nginx proxies to them).

A6. Clean Docker Space Before Build

```
# Check Docker disk usage
docker system df

# Remove old/dangling images (safe — only removes unused)
docker image prune -f

# More aggressive cleanup (removes all unused images)
docker image prune -a -f

# Check space after cleanup
df -h /
```

Part B: Deploy to DEV

B1. Pull Latest Code

```
cd /opt/global-erp
git fetch origin
git pull origin main
```

> If you get ownership errors: `git config --global --add safe.directory /opt/global-erp`

B2. Backup DEV Database

```
mkdir -p /opt/backups

docker compose -p global-erp-dev -f docker-compose.dev.yml exec postgres \
  pg_dump -U autoguru global_erp_dev > /opt/backups/dev-$(date +%Y%m%d-%H%M%S).sql

# Verify backup
ls -lh /opt/backups/dev-*.sql | tail -1
```

B3. Rebuild & Start DEV

```
docker compose -p global-erp-dev -f docker-compose.dev.yml down
docker compose -p global-erp-dev -f docker-compose.dev.yml build --no-cache
docker compose -p global-erp-dev -f docker-compose.dev.yml up -d
```

B4. Run Migrations on DEV

```
docker compose -p global-erp-dev -f docker-compose.dev.yml exec web pnpm
db:migrate
```

Verify:

```
docker compose -p global-erp-dev -f docker-compose.dev.yml exec postgres \
  psql -U autoguru -d global_erp_dev -c "SELECT * FROM schema_migrations ORDER BY
id DESC LIMIT 15;"
```

Should show up to `186_rcc_permissions`.

B5. Verify DEV

```
docker compose -p global-erp-dev -f docker-compose.dev.yml ps
docker compose -p global-erp-dev -f docker-compose.dev.yml logs web --tail=30
curl -s -o /dev/null -w "%{http_code}" http://127.0.0.1:3001
```

B6. DEV Smoke Tests

Open `https://dev.yourdomain.com`:

- [ ] Login works
- [ ] Sidebar shows **Revenue Command Center**
- [ ] Sidebar shows **Sales Center** (not "Call Center")
- [ ] PIS → Advisor Portal loads
- [ ] Inspection → Estimate → Approval flow
- [ ] Invoice creation + payment
- [ ] Vendor → Part details + bids
- [ ] AI Panel responds (multi-provider)

Part C: Deploy to PROD

> **Only after DEV smoke tests pass.**

C1. Backup PROD Database (CRITICAL)

```
docker compose -p global-erp-prod -f docker-compose.prod.yml exec postgres \
  pg_dump -U autoguru global_erp_prod > /opt/backups/prod-$(date +%Y%m%d-
%H%M%S).sql

# Verify backup exists and has real data
ls -lh /opt/backups/prod-*.sql | tail -1
```

C2. Rebuild & Start PROD

```
docker compose -p global-erp-prod -f docker-compose.prod.yml down
docker compose -p global-erp-prod -f docker-compose.prod.yml build --no-cache
docker compose -p global-erp-prod -f docker-compose.prod.yml up -d
```

> **Downtime:** ~3-8 minutes (Next.js prod build). Plan accordingly.

C3. Run Migrations on PROD

```
docker compose -p global-erp-prod -f docker-compose.prod.yml exec web pnpm
db:migrate
```

Verify:

```
docker compose -p global-erp-prod -f docker-compose.prod.yml exec postgres \
  psql -U autoguru -d global_erp_prod -c "SELECT * FROM schema_migrations ORDER BY
id DESC LIMIT 15;"
```

C4. Verify PROD

```
docker compose -p global-erp-prod -f docker-compose.prod.yml ps
docker compose -p global-erp-prod -f docker-compose.prod.yml logs web --tail=30
curl -s -o /dev/null -w "%{http_code}" http://127.0.0.1:3000
curl -s -o /dev/null -w "%{http_code}" https://yourdomain.com
```

C5. PROD Smoke Tests

Open `https://yourdomain.com`:

- ☐ Login with real credentials
- ☐ Revenue Command Center accessible + data loads
- ☐ Sales Center + WebSocket connection
- ☐ PIS Advisor Portal functional
- ☐ Full flow: Inspection → Estimate → Approval → Invoice → Payment
- ☐ Vendor bids + part details
- ☐ AI features respond
- ☐ No console errors

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6. Post-Deployment Checks

```
# All containers running
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"

# Resource usage (ensure no OOM)
docker stats --no-stream

# Disk space
df -h /

# Nginx healthy
sudo nginx -t && sudo systemctl status nginx

# Monitor prod logs for 5-10 mins
docker compose -p global-erp-prod -f docker-compose.prod.yml logs -f web
```

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7. Rollback Plan

PROD Rollback Steps:

```
# 1. Stop prod
docker compose -p global-erp-prod -f docker-compose.prod.yml down

# 2. Go back to last known good commit
cd /opt/global-erp
git log --oneline -5
git checkout <good-commit>

# 3. Rebuild + start
docker compose -p global-erp-prod -f docker-compose.prod.yml build --no-cache
docker compose -p global-erp-prod -f docker-compose.prod.yml up -d

# 4. Restore database if migrations broke something
docker compose -p global-erp-prod -f docker-compose.prod.yml exec -T postgres \
  psql -U autoguru -d global_erp_prod < /opt/backups/prod-backup-YYYYMMDD-
HHMMSS.sql
```

> Migrations are forward-only — restore from backup if needed.

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8. Quick Copy-Paste Commands

Full Deploy Sequence (SSH into EC2, then run):

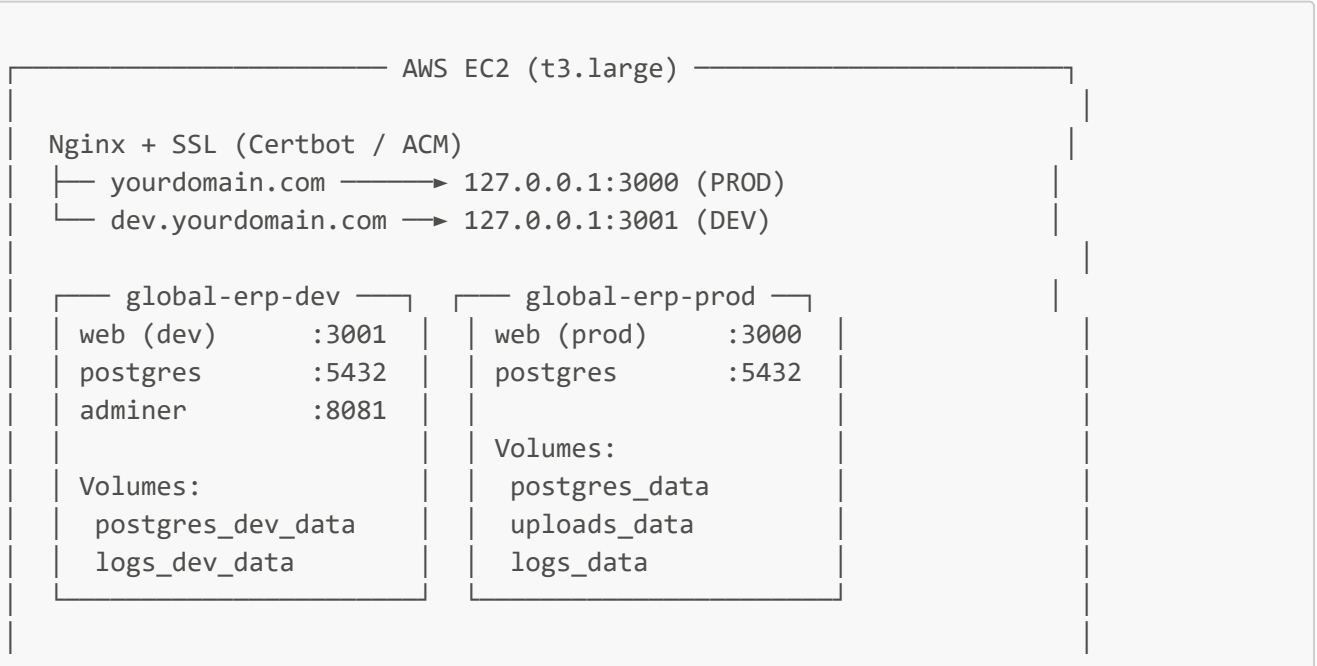
```
# ===== FULL DEPLOY — COPY & RUN STEP BY STEP =====

cd /opt/global-erp
git pull origin main
mkdir -p /opt/backups

# ── DEV ──────────────────────────────────────────
docker compose -p global-erp-dev -f docker-compose.dev.yml exec postgres \
  pg_dump -U autoguru global_erp_dev > /opt/backups/dev-$(date +%Y%m%d-%H%M%S).sql

docker compose -p global-erp-dev -f docker-compose.dev.yml down
docker compose -p global-erp-dev -f docker-compose.dev.yml build --no-cache
docker compose -p global-erp-dev -f docker-compose.dev.yml up -d
docker compose -p global-erp-dev -f docker-compose.dev.yml exec web pnpm
db:migrate
docker compose -p global-erp-dev -f docker-compose.dev.yml ps

echo ">>> DEV deployed. Run smoke tests on dev.yourdomain.com <<<"

# ── PROD (run after DEV smoke tests pass) ────────
docker compose -p global-erp-prod -f docker-compose.prod.yml exec postgres \
  pg_dump -U autoguru global_erp_prod > /opt/backups/prod-$(date +%Y%m%d-
```

```
%H%M%S).sql

docker compose -p global-erp-prod -f docker-compose.prod.yml down
docker compose -p global-erp-prod -f docker-compose.prod.yml build --no-cache
docker compose -p global-erp-prod -f docker-compose.prod.yml up -d
docker compose -p global-erp-prod -f docker-compose.prod.yml exec web pnpm
db:migrate
docker compose -p global-erp-prod -f docker-compose.prod.yml ps

echo ">>> PROD deployed. Run smoke tests on yourdomain.com <<<"
```

Useful Aliases (add to `~/.bashrc`):

```
alias dev-up="cd /opt/global-erp && docker compose -p global-erp-dev -f docker-
compose.dev.yml up -d"
alias dev-down="cd /opt/global-erp && docker compose -p global-erp-dev -f docker-
compose.dev.yml down"
alias dev-logs="cd /opt/global-erp && docker compose -p global-erp-dev -f docker-
compose.dev.yml logs -f web"
alias dev-migrate="cd /opt/global-erp && docker compose -p global-erp-dev -f
docker-compose.dev.yml exec web pnpm db:migrate"
alias prod-up="cd /opt/global-erp && docker compose -p global-erp-prod -f docker-
compose.prod.yml up -d"
alias prod-down="cd /opt/global-erp && docker compose -p global-erp-prod -f
docker-compose.prod.yml down"
alias prod-logs="cd /opt/global-erp && docker compose -p global-erp-prod -f
docker-compose.prod.yml logs -f web"
alias prod-migrate="cd /opt/global-erp && docker compose -p global-erp-prod -f
docker-compose.prod.yml exec web pnpm db:migrate"
```

Architecture



```
  │  Security Group: 22 (SSH), 80, 443 only                              │
  │  Elastic IP recommended (keeps IP after stop/start)                  │
  └──────────────────────────────────────────────────────────────────────┘
```

Prepared: 2026-03-31 | **Applies to:** Commits `42f3732` → `93ea99a` (12 commits, 14 migrations)